بيانك
Bayanuk

AI Stack Decision

API-based models for contract analysis, chat, semantic search, and OCR fallback in the Bayanuk MVP.

Product	Bayanuk (بيانك), AI review of job offers and employment contracts
Owner	JX Tech Studio
Version	2.0 (replaces v1.0, the self-hosted plan)
Date	October 2026

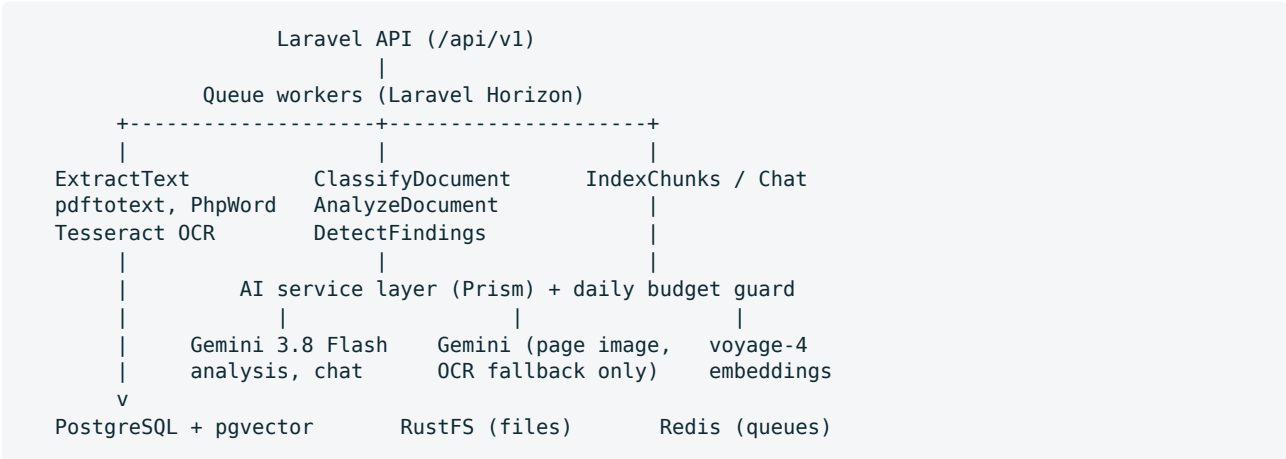
1. Decision summary

Bayanuk uses paid AI APIs instead of hosting models on its own server. The goal is the lowest running cost with high quality in Arabic and English, and no GPU server to buy or run. Every provider sits behind the AIService interface, so any one of them can be replaced later without rebuilding the product.

This version replaces v1.0 (Qwen3-8B and BGE-M3 on a self-hosted GPU server).

Layer	Choice	Purpose
Analysis and chat	Gemini 3.8 Flash (Google Gemini API, paid tier)	Classification, extraction, attention points, clause explanations, and answers grounded in the contract.
Embeddings	voyage-4 (Voyage AI API), 1024 dimensions	Embeds contract chunks and questions for retrieval.
Text extraction	pdftotext (Poppler), PhpWord for DOCX	Text per page from text-based PDFs and Word files. Free, runs locally.
OCR	Tesseract (Arabic + English), then Gemini for unreadable pages	Scanned pages. The API is used only when local OCR fails.
Attention points	Deterministic PHP rules, plus AI clause detection	Rules on extracted terms; AI only for clauses rules cannot detect.
Data and vectors	PostgreSQL + pgvector, RustFS for files	All data, chunks, and embeddings stored on our server.
Integration	Prism behind our AIService interface	One interface for all AI tasks; providers and models are configuration values.

2. Architecture



No AI model runs on our server. Files, extracted text, results, and embeddings stay on our server; only the text each task needs is sent to the providers.

3. Components

3.1 Contract analysis and chat: Gemini 3.8 Flash

- Analyze employment contracts and job offers in Arabic and English
- Extract salary, leave, probation, notice, and termination terms
- Detect clauses that need attention
- Explain contract clauses to the user
- Answer user questions based on the contract text
- Return structured JSON with quotes from the contract

3.2 Staged analysis

The model does not reread the full contract for every task. Each stage sends only what it needs:

Stage	What is sent	Output
Classify	First pages only	Document type and language.
Extract	Full contract text, once	Key terms with page and quote, and the summary.
Detect findings	Extracted terms; relevant chunks for AI clause checks	Attention points. Rules run first, at no cost.
Chat	Retrieved chunks only (top 6)	Answer with page citations.

No reranking model at the start. Retrieval quality is tested first, and a reranker is added only if tests show it is needed.

3.3 Semantic search: voyage-4

- Embeds contract chunks (about 500 tokens, never across pages) and user questions
- 1024 dimensions; database column `vector(1024)`; within the 2,000-dimension HNSW limit
- Multilingual model, chosen over voyage-4-lite because Voyage documents multilingual support for voyage-4
- Retrieval keeps the full contract out of every chat request

3.4 Document processing

- **Text PDFs:** pdftotext from Poppler, text per page, no API call
- **DOCX:** PhpWord, split into logical pages
- **Scanned or broken pages:** Tesseract (ara + eng) first. If the page is still unreadable, the page image is sent to Gemini, and the page is saved with `extraction_method = vision`
- Personal identifiers are masked before text is sent to any provider

4. Cost

4.1 Prices

Model	Price per 1M tokens	Notes
Gemini 3.8 Flash	$0.75 input / $3.75 output	Until December 31, 2026.
Gemini 3.8 Flash	$1.50 input / $7.50 output	From January 1, 2027.
voyage-4	$0.06	The first 200 million tokens are free.
pdftotext, Tesseract	Free	Run on our server.

Prices and model names must be checked again before launch, since they can change.

4.2 Estimate per analysis

A typical contract: about 18,000 input tokens and 2,000 output tokens.

Period	Per analysis	Per 1,000 analyses
Until December 2026	about $0.021	about $21
From January 2027	about $0.042	about $42

Real cost depends on contract length, the number of stages, chat questions, retries, and OCR fallback pages.

4.3 Daily budget

> **Daily AI budget: $2.** Stored as `BAYANUK_DAILY_AI_BUDGET_USD` and checked by the application before every AI call, using the cost saved in `usage_events`. Provider spending alerts are not relied on as a hard stop. At current prices, $2 covers about 95 analyses a day, and about 48 from January 2027. When the budget is reached, new analyses pause with a clear message until the next day. The budget is reviewed after the first 50 real analyses, as the MVP Definition requires.

4.4 Cost reduction

- Never send the full contract with every question
- Store chunks and embeddings; retrieve only the relevant chunks
- Split the analysis into stages (section 3.2)
- No reranking model unless tests prove it is needed
- Track tokens and cost per contract and per user
- Limit retries; reuse the existing analysis when the same file is uploaded again

5. Privacy and security

- API keys are kept in server environment variables only, never in the repository; separate keys for staging and production

- Send providers only the data each task needs
- Gemini is used on the paid tier. Under Google's published terms, paid-tier prompts and responses are not used to improve Google's products. This does not mean the data is not processed by the provider
- Review each provider's data processing, retention, and legal terms before sending real contracts
- Processing happens outside Saudi Arabia; check the requirements of the Personal Data Protection Law (PDPL) and state the providers in the privacy policy
- Masking names, phone numbers, and IBANs is not enough to anonymize the contract owner; the contract text itself can still identify them
- Bayanuk's deletion policy applies to all files and data stored on our side
- Development and testing use sample or anonymized contracts only, never real ones

6. Testing before launch

Area	Target or check
Extraction accuracy	90% on core fields, on Arabic, English, and mixed contracts (MVP).
Grounding	Every result links to real text in the contract; quotes are verified.
Retrieval quality	Correct page in the top results for golden questions; decides if a reranker is needed.
Time to result	Under 60 seconds for a 10-page text PDF, 90th percentile (MVP).
Chat answer time	Under 8 seconds (MVP).
Large and scanned files	30-page files, scanned files, and broken Arabic text.
User isolation	No user or guest can reach another owner's data.
Cost	Cost per analysis and per chat, per user, matches the estimate.

7. Changes to existing documents

Document	Section	Change
Database Schema	3.5, 3.6, 3.13	embedding_model is voyage-4; provider is gemini; cost_usd holds the real cost again.
Database Schema	3.4	vision is used for pages read by Gemini after OCR fails.
Technical Plan	3, 6	Gemini 3.8 Flash and voyage-4; staged analysis; no reranker at the start.
Technical Plan	6.5, 11	Daily AI budget of $2 checked in the application.

8. Risks and open points

Item	Plan
Backup provider	Open. Prism allows a second provider or model by configuration; the choice is not made yet.
Gemini price increase in January 2027	Cost per analysis roughly doubles; the $2 budget covers about half as many analyses.
Provider outage	Queue retries with backoff; clear "try again later" message; Grafana alert.
Arabic quality	Golden set decides before launch; models can be changed by configuration.
Budget reached early	New analyses pause until the next day; review the budget and limits.
Data transfer outside Saudi Arabia	Legal review before real contracts; stated in the privacy policy.